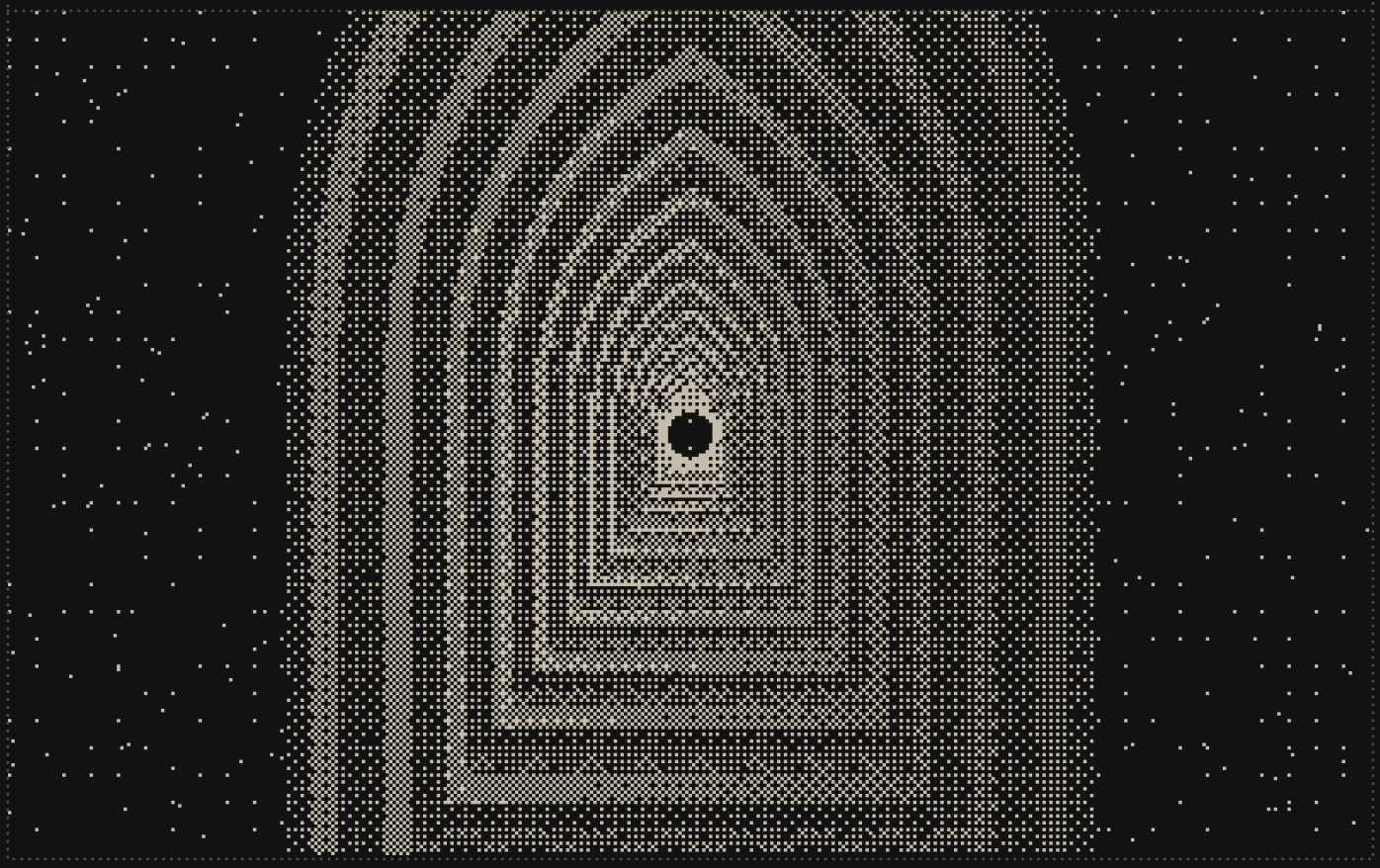
PLATE // ZENO'S NAVE



KAELEARN // ANALYSIS I // FOLIO 03

# Cauchy Sequences

*How to prove a sequence arrives without knowing where it is going.*

| | |
|---|---|
| FOLIO | 03 |
| SUBJECT | Analysis I |
| SOURCES | Specimen: no real sources |
| READ | 2 min view · 14 min folio |
| FILED | 2026-10-11 |



PART I

# The trouble with limits

To prove that a sequence converges, the definition you were given makes a strange demand: name the limit first. In

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N : \; |a_n - L| < \varepsilon$$

the $L$ sits right in the middle, and you are expected to bring it. For $a_n = 1/n$ that costs nothing. For $a_n = \sum_{k=1}^{n} 1/k^2$ you would need to know the answer is $\pi^2/6$, which took Euler years. Most sequences do not arrive with their limit printed on the box.

Cauchy's move: stop asking *where* the terms are going. Ask only whether they are getting close to *each other*. That you can check with nothing but the terms themselves.

*Cauchy wrote it down in 1821. Bolzano had it four years earlier, in Prague, and nobody read him. I read him.*

THE TWO-MINUTE VIEW // READ THIS FIRST

- **The problem.** The $\varepsilon$–$N$ definition needs the limit $L$ in advance.
- **The fix.** A Cauchy sequence *(Cauchy-Folge)*: past some point, all terms huddle, $\forall \varepsilon > 0 \; \exists N \; \forall m, n \geq N : |a_m - a_n| < \varepsilon$.
- **Convergent ⇒ Cauchy**, always, by one triangle inequality.
- **Cauchy ⇒ convergent** holds in $\mathbb{R}$ and fails in $\mathbb{Q}$. This *is* completeness *(Vollständigkeit)*.
- **The proof** runs Cauchy ⇒ bounded ⇒ a convergent subsequence (Bolzano–Weierstrass) ⇒ the whole sequence follows it.
- **The trap.** $|a_{n+1} - a_n| \to 0$ is not enough: the harmonic sums take ever smaller steps and still walk to infinity.

GUESS FIRST // WRONG IS FINE, GUESSING IS THE POINT

01 If consecutive terms get arbitrarily close, $|a_{n+1} - a_n| \to 0$, must the sequence converge?

02 The decimal truncations of $\sqrt{2}$, namely $1,\ 1.4,\ 1.41,\ 1.414, \ldots$, all live in $\mathbb{Q}$. Do they converge *in* $\mathbb{Q}$?

03 Can you prove $\sum 1/k^2$ converges without knowing its value? How would you even start?

## Terms · *Begriffe*

| ENGLISH | DEUTSCH | IN ONE LINE |
|---|---|---|
| Cauchy sequence | *Cauchy-Folge* | all terms past $N$ within $\varepsilon$ of each other |
| complete | *vollständig* | every Cauchy sequence converges |
| subsequence | *Teilfolge* | $(a_{n_k})$ with $n_1 < n_2 < \cdots$ |
| bounded | *beschränkt* | $\lvert a_n \rvert \le M$ for all $n$ |
| accumulation point | *Häufungspunkt* | the limit of some subsequence |
| triangle inequality | *Dreiecksungleichung* | $\lvert x+y \rvert \le \lvert x \rvert + \lvert y \rvert$ |

PART II

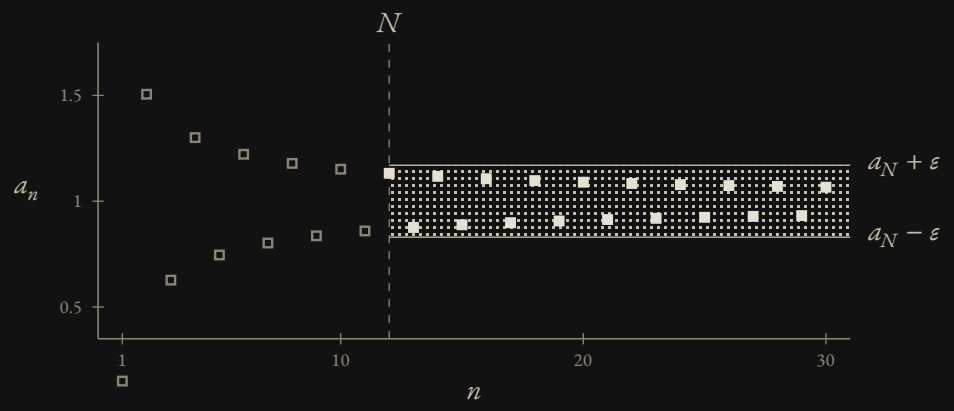
# The definition

DEFINITION 03.1 // CAUCHY SEQUENCE · CAUCHY-FOLGE

A sequence $(a_n)$ of real numbers is a *Cauchy sequence* if

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall m, n \ge N : \quad \lvert a_m - a_n \rvert < \varepsilon.$$

Read the quantifiers in order, because the order is the content. First someone hands you an $\varepsilon$, as small as they like. Then you choose $N$, and $N$ may depend on $\varepsilon$. Then *every* pair of terms past $N$ must sit within $\varepsilon$ of each other. Not neighbours. Every pair: the thousandth and the $10^{100}$th.




*Past $N$ every term stays within $\varepsilon$ of $a_N$, so any two are within $2\varepsilon$ of each other. The band is centred on a term, not on the limit. You never needed $L$.*

THEOREM 03.2 // THE CAUCHY CRITERION · CAUCHY-KRITERIUM

A sequence of real numbers converges if and only if it is a Cauchy sequence.

PROOF IN ONE BREATH $\Rightarrow$: two terms near $L$ are near each other, $|a_m - a_n| \le |a_m - L| + |L - a_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$. $\Leftarrow$: Cauchy $\Rightarrow$ bounded $\Rightarrow$ Bolzano–Weierstrass hands you a subsequence with a limit $L \Rightarrow$ the huddle drags every term to $L$.

### *Why does the $\Leftarrow$ direction need Bolzano–Weierstrass?*

Being Cauchy says nothing about *where* the terms are, only that they huddle. Something has to produce a candidate $L$, a point to huddle around. Bolzano–Weierstrass is that something, and it is exactly where the completeness of $\mathbb{R}$ enters. In $\mathbb{Q}$ the huddle can form around a hole: the decimals of $\sqrt{2}$ huddle, and $\mathbb{Q}$ has nothing at the centre.

Why?

### *Why is $|a_{n+1} - a_n| \to 0$ not enough?*

It controls neighbours, not distant pairs. The harmonic sums $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ move by $\frac{1}{n+1} \to 0$ per step, yet $H_{2n} - H_n \ge \frac{1}{2}$ forever. Tiny steps, infinite walk.

Why?

*If you keep one sentence: Cauchy trades a destination for a crowd.*

## TRIGGERS // IF YOU SEE THIS, REACH FOR THAT

| YOU SEE | REACH FOR |
|---|---|
| Convergence to prove, no candidate limit | the Cauchy criterion |
| $\lvert a_m - a_n \rvert$ with both indices free | let $m > n$, telescope: $\sum_{k=n}^{m-1} \lvert a_{k+1} - a_k \rvert$ |
| $\lvert a_{n+1} - a_n \rvert \le q \lvert a_n - a_{n-1} \rvert$, $q < 1$ | a geometric tail, $\le \frac{q^n}{1-q} \lvert a_1 - a_0 \rvert$ |
| A series $\sum b_k$ | Cauchy for partial sums: $\lvert \sum_{k=n+1}^{m} b_k \rvert < \varepsilon$ |



## FOR THE DECK // MAKE THESE YOURSELF, THE MAKING IS THE LEARNING

| | | |
|---|---|---|
| DEF | Cauchy sequence: the definition, quantifiers in order. | $\forall \varepsilon > 0 \; \exists N \; \forall m, n \ge N : \; \lvert a_m - a_n \rvert < \varepsilon$ |
| CLOZE | In {{$\mathbb{R}$}}, a sequence converges iff it is {{Cauchy}}. | The first gap is the hypothesis that fails in $\mathbb{Q}$. |
| WHY | Why does Cauchy $\Rightarrow$ convergent fail in $\mathbb{Q}$? | The decimals of $\sqrt{2}$ are Cauchy in $\mathbb{Q}$; the point they huddle round is missing. |
| TRIGGER | Convergence to prove, limit unknown. Reach for? | The Cauchy criterion, usually via a telescoping or contraction bound. |